Change Order: Amazon API & Certification Data Enhancements

GreenCircle Certified Database

Prepared For: GreenCircle Certified, LLC
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Status: Proposed (Updated)

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1. Background

Amazon's Climate Pledge Friendly (CPF) program is requesting richer certification data. Rather than a binary "this product is certified," Amazon wants quantitative values such as "*This product uses 32% less energy than average*" and the ability to filter CPF eligibility using actual percentage thresholds.

> **Confirmed by Amazon (Kyle):**
>
> Amazon will use the `certificationResults` field for two purposes:
>
> 1. **Customer-facing display**
>
>   — showing quantitative results (e.g., energy saving %, recycled content %) on product pages
> 2. **Internal CPF filtering**
>
>   — moving from binary eligibility to threshold-based filtering so Amazon can model the impact of changes before committing them

> **Confirmed by GreenCircle (Carly):**
>
> A Pathway field with admin-managed dropdown values is needed to provide context for certification results. For SEP certifications, the same percentage may relate to different methodologies (e.g., Design Efficiency vs. Consumer Behavior). The approach must avoid cert-type-specific fields since "*things are always changing.*"

This change order proposes the work required to meet these requirements while respecting GreenCircle's directive to keep the system simple and avoid cert-type-specific fields.

2. What Already Exists

Capability	Status
Certification Results field (free-text, 1,000 chars) on every certificate	BUILT
Certification Results returned in API responses (standard JSON and GS1 JSON-LD)	BUILT

Capability	Status
Certification Type + Subcategory hierarchy (admin-managed)	BUILT
Amazon Code mapping on each Certification Type	BUILT
Product Identifiers (ASIN, UPC, GTIN, EAN) with partner tagging	BUILT
GS1 JSON-LD API endpoint for Amazon integration	BUILT
Partner API key management with automatic filtering	BUILT
Certificate PDF download via API	BUILT

Gaps

Gap	Impact
Amazon Code exists in database but not returned in API responses	Amazon cannot use it for CPF mapping
Pathway concept does not exist	No way to categorize the methodology behind certification results (confirmed needed by GreenCircle for SEP)

3. Proposed Changes

#	Change	Details	Risk	DB Change
1	Expose Amazon Code in API responses	Return `amazonCode` (e.g., `greencircle_energy_savings`) in both the standard JSON and GS1 JSON-LD endpoints so Amazon can map certificates to CPF categories	LOW	NO
2	Add Pathway field	Admin-managed lookup table, dropdown on certificate form scoped by cert type, exposed in API responses. Pending: initial values — see Section 5	MEDIUM	1 TABLE
3	Expose Pathway in API responses	Return `pathway` value in both the standard JSON and GS1 JSON-LD endpoints	LOW	NO
4	Soft validation warning for Amazon certificates	When Partner = Amazon and Certification Results is empty, display a warning (not a blocker) to the analyst. Pending: confirmation — see Section 5	LOW	NO

#	Change	Details	Risk	DB Change
5	Updated API documentation	Refresh Swagger documentation and Amazon integration guide with new fields	LOW	NO

> **Summary:**
> 5 changes total. 1 database migration (new Pathway table). All other changes are additive with no schema impact. All new API fields are optional — existing consumers are unaffected.

4. Pathway Design: Admin-Managed Lookup Table

GreenCircle has confirmed the approach: an admin-managed dropdown where each value can optionally be scoped to a specific Certification Type.

Example Pathway	Scoped To	Available For
(values TBD — see Section 5)	SEP / Energy Savings	Only SEP certs
General	*(all types)*	Any certificate

How it works

- Analyst selects Certification Type → Pathway dropdown shows only relevant options
- Pathway remains optional; dropdown hidden if no pathways configured for the cert type
- Admin manages values without developer involvement
- Prevents free-text inconsistency that would degrade API quality for Amazon

Why this approach

- Data consistency is high (controlled values, no variant spellings)
- Admin has full control to add/modify/remove pathway values as cert types evolve
- Clean API filterability for Amazon
- Future-proof: new cert types get their own pathways without code changes
- Not cert-type-specific — the same generic Pathway mechanism serves all cert types

5. Confirmed & Pending Items

Confirmed by Email Thread (Feb 5, 2026)

Item	Confirmed By	Detail
Amazon will use `certificationResults`	Kyle (Amazon)	For customer-facing display and internal CPF threshold filtering
`certificationResults` already available in API	Development	Returned in both standard JSON and GS1 JSON-LD endpoints
Amazon wants energy saving % and recycled content %	Kyle (Amazon)	Specific use cases mentioned
Amazon is fine with partner-filtered API	Kyle (Amazon)	Only sees Amazon-tagged certificates
Pathway field is needed	Carly (GreenCircle)	For SEP certs, need to know which path the number relates to
Pathway should be admin-managed dropdown	Carly (GreenCircle)	"Pathway field with dropdowns (that Admin can control)"
Avoid cert-type-specific fields	Carly (GreenCircle)	"Things are always changing" — generic approach preferred

Pending — Answers Needed to Complete Changes #2–4

> **Change #1 and #5 can proceed immediately.**
>
> Changes #2–4 require the answers below.

#	Question	For	Blocks
1	What are the initial Pathway values for SEP? (e.g., "Design Efficiency", "Consumer Behavior", "Operational Improvement"?)	GreenCircle	#2
2	Are there other cert types besides SEP that need pathway values right away?	GreenCircle	#2
3	Should Pathway be visible to external users, or internal/API-only?	GreenCircle	#2
4	Confirm: when Partner = Amazon and Cert Results is empty, display a soft warning to the analyst?	GreenCircle	#4

6. What Is NOT Included

Excluded Item	Reason
Cert-type-specific fields	GreenCircle directive: "*avoid creating multiple fields or fields specific to a certain cert type*"
Mandatory Certification Results for all certs	Only Amazon-tagged certs may need this
Forced numeric validation on Certification Results	Results vary by type: percentages, levels, scores. Forcing numeric breaks legitimate data.
New API version (v2)	Adding optional fields to v1 is non-breaking
Changes to the 14-state certificate workflow	Unrelated to Amazon's request

7. Risk Assessment

Risk	Severity	Likelihood	Mitigation
Pathway design does not fit future cert types	MEDIUM	LOW	Option A scoping + "General" fallback; admin can adjust without developer
Validation blocks analyst workflows	HIGH	MEDIUM	Warnings only, never hard blockers
Existing certificate data incompatible	LOW	LOW	All new fields optional; existing records unaffected
API changes break current integration	HIGH	LOW	All additions are new optional fields; no removals

8. API Fields After This Change Order

The API will return the following fields per certificate. New fields are highlighted.

Field	Description	Status
Certificate Number	Unique certificate identifier (e.g., GCC-2026-001)	EXISTING
Status	Certificate status (e.g., Active, Expired)	EXISTING
Issue Date / Expiration Date	Validity period	EXISTING

Field	Description	Status
Company	Company name and ID	EXISTING
Product	Product name and ID	EXISTING
Location / Country	Where the certification applies	EXISTING
Partner	Associated partner (e.g., Amazon)	EXISTING
Certification Type	Type and subcategory (e.g., Energy Savings)	EXISTING
Certification Results	Quantitative result (e.g., "32% energy savings")	EXISTING
Product Identifiers	ASIN, UPC, GTIN, EAN values	EXISTING
PDF Download	Link to download certificate PDF	EXISTING
Amazon Code	CPF mapping code (e.g., `greencircle_energy_savings`)	NEW
Pathway	Methodology context for certification results (e.g., Design Efficiency)	NEW